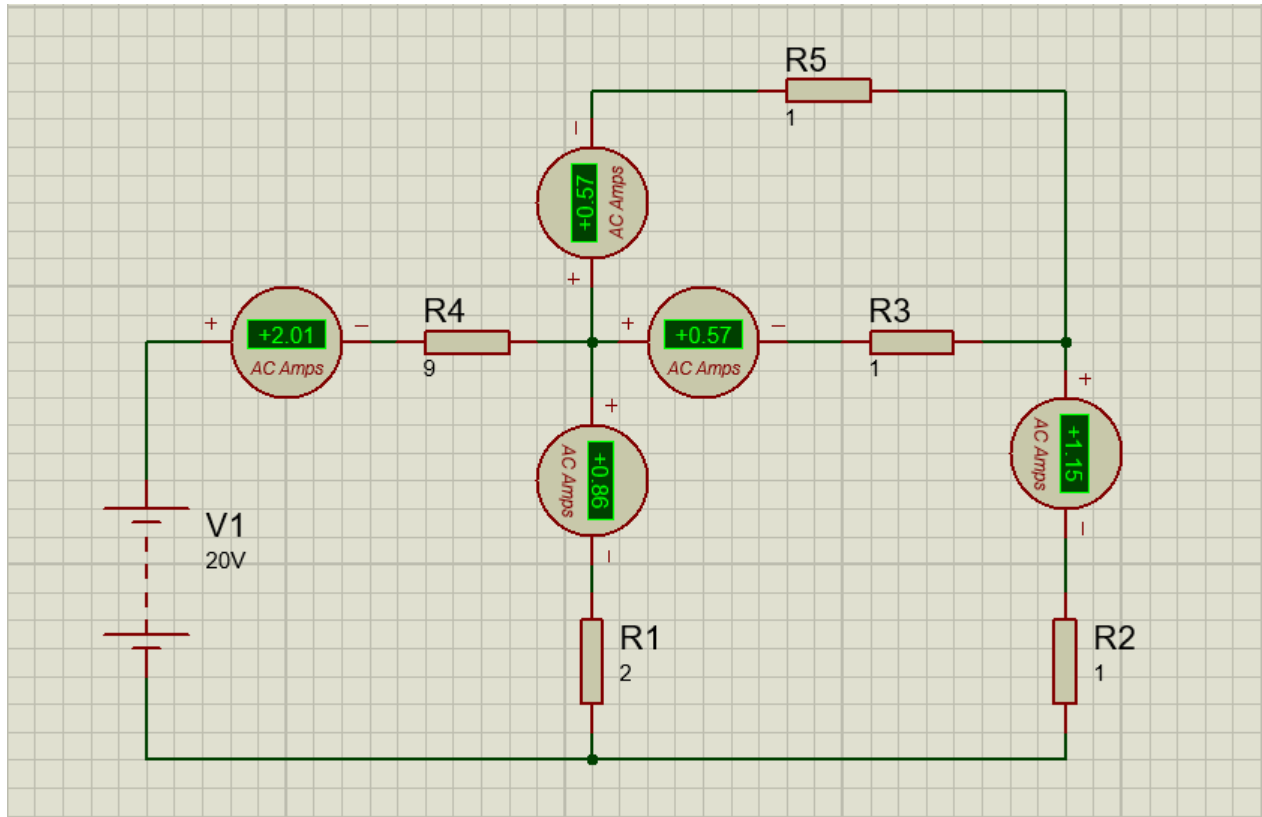


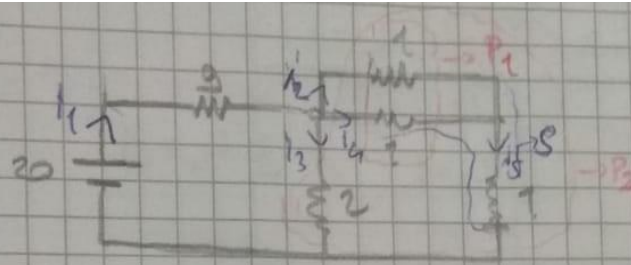
SORU 1 :



Tablo

	i_4	i_2	i_3	i_4	i_5
Manuel	2,028	0,579	0,967	0,579	1,158
Simulasyon	2,01	0,57	0,966	0,57	1,15

İŞLEMLEMLER :



$$R \Rightarrow P_1 = \frac{1}{1} + \frac{1}{2} = \frac{1}{2}$$

$$p_2 = \frac{2}{3} + \frac{1}{2} = \frac{2}{6} + \frac{3}{6} = \frac{5}{6}$$

$$\frac{1}{2} + 1 = \frac{3}{2}$$

$$R_{E2} = 6\frac{1}{7} + 9 = \frac{69}{7}$$

$$I' \Rightarrow \textcircled{1} \quad I' = \frac{V}{R} = \frac{20}{\frac{69}{7}} = \frac{20 \cdot 7}{69} = 2,028$$

$$I_1 = I_2 + I_3 + I_4$$

$$I_2 = I_5 + I_3$$

$$1/5 = 1/4 + 1/20$$

(2) $\frac{3}{2} \cdot 15 = 2.13$

$$7k = 2,028$$
$$k = 289$$

$$\begin{aligned} i_5 &= 1,158 \\ i_3 &= 0,867 \end{aligned}$$

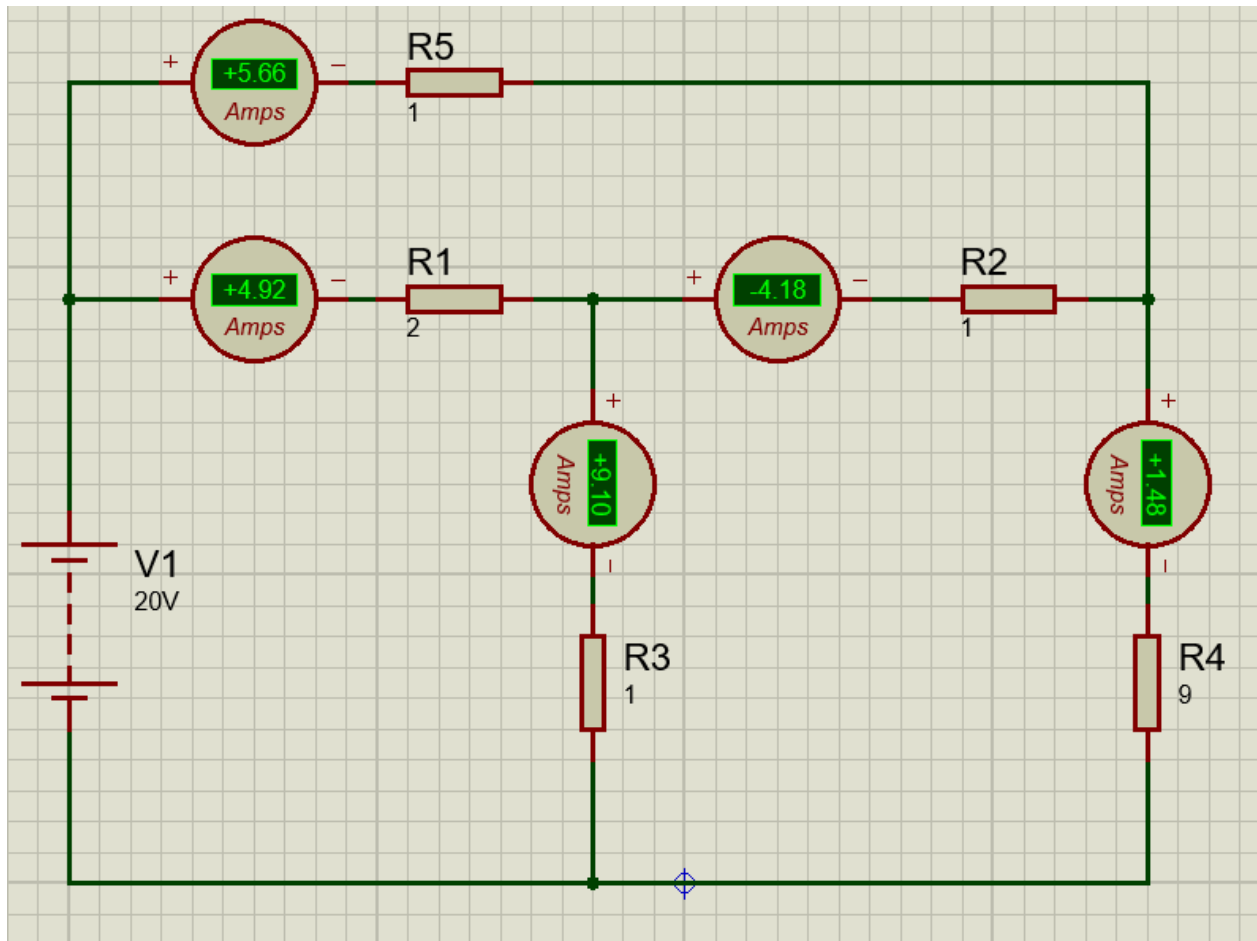
$$\begin{array}{ccc} 3 & & 4 \\ \downarrow & & \downarrow \\ 2 & & 3 \\ 4 & & 4 \end{array}$$

③ $1 \cdot \frac{1}{3} = \frac{1}{3} \cdot 1$

$$\frac{1}{16} + \frac{1}{9} = \frac{1}{5} = 2,158$$

$$1_4 = 1_3 = 0,579$$

SORU2:

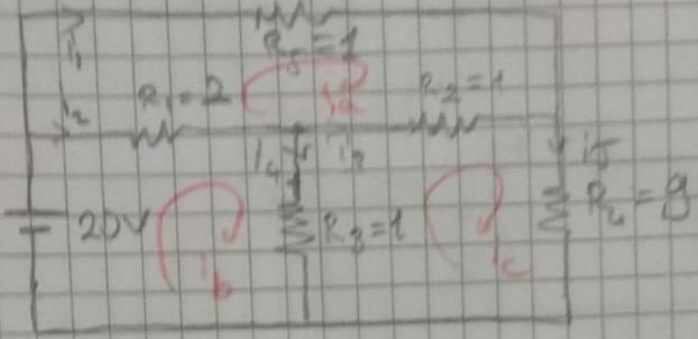


TABLO2:

Date :/...../.....

	i_1	i_2	i_3	i_4	i_5
Manuel	4,6	4	-3,4	7,4	1,2
Simulasyon	5,6	4,92	-4,18	9,10	1,48

İŞLEMLER2:



$i \rightarrow$

$$\begin{aligned} (1) \rightarrow 20 + 2(i_b - i_a) + i_b - i_c &= 0 \rightarrow -i_a + 3i_b - i_c = 20 \\ (2) \rightarrow 2(i_a - i_b) + i_a + i_b - i_c &= 0 \rightarrow 4i_a - 2i_b - i_c = 0 \\ (3) \rightarrow i_c - i_b + i_c - i_a + 8i_c &= 0 \rightarrow -i_a - i_b + 11i_c = 0 \end{aligned}$$

$$\downarrow$$

$$\begin{pmatrix} -1 & 3 & -1 \\ 4 & -2 & -1 \\ -1 & -1 & 11 \end{pmatrix} \begin{pmatrix} i_b \\ i_a \\ i_c \end{pmatrix} = \begin{pmatrix} 20 \\ 0 \\ 0 \end{pmatrix}$$

$$i_a = \frac{23}{5} = 4,6$$

$$i_b = \frac{47}{5} = 9,4$$

$$i_c = \frac{6}{5} = 1,2$$

$$\begin{aligned} i_1 = i_a &= 4,6 \\ i_2 = i_b - i_a &= 4 \\ i_3 = i_c - i_a &= -3,4 \\ i_4 = i_b - i_c &= 7,4 \\ i_5 = i_c &= 1,2 \end{aligned}$$

AÇIKLAMA :

Değerler arasındaki farkın sebebi özdirençlerin hesaba katılmamasıdır.